

datalex



Introduction to Big Data



Alexandre BERGERE

Head of Data & AI Engineer & Partners at DataGalaxy, Data Architect independent
Delta & openLineage lover

Data Expert
(Audit, Modeling, Urbanization, BI's project – Azure, Big Data, Spark, openLineage & Delta lover)

Trainer
(BI, Big Data, MongoDB, NoSQL, SQL, Cloud Computing)

Speaker
(Pass Summit 2018, Spark AI Tour 2020, …)

Cloud & data tech articles
(medium, LinkedIn, slideshare)

Avanade
Sr Anls, Data Analyst

2016 - 2019

Trainer
2016 - x

ESAIP, ESME, ECS, NEOMA, ESEO, EFREI, Sorbonne Paris

BI, Big Data, MongoDB, NoSQL, Panorama de la data, Cloud computing

OpenClassroom
Mentor

2019 - x

Datalex
2019 - x

Data Architect Freelance

DataRedKite
CTO & Co-Founder

2020 - 2023

DataGalaxy
2021 - x

Head of Data & AI Engineer

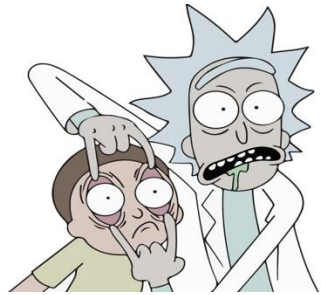
Don’t get lost



Practice



Demo



Important

Summary

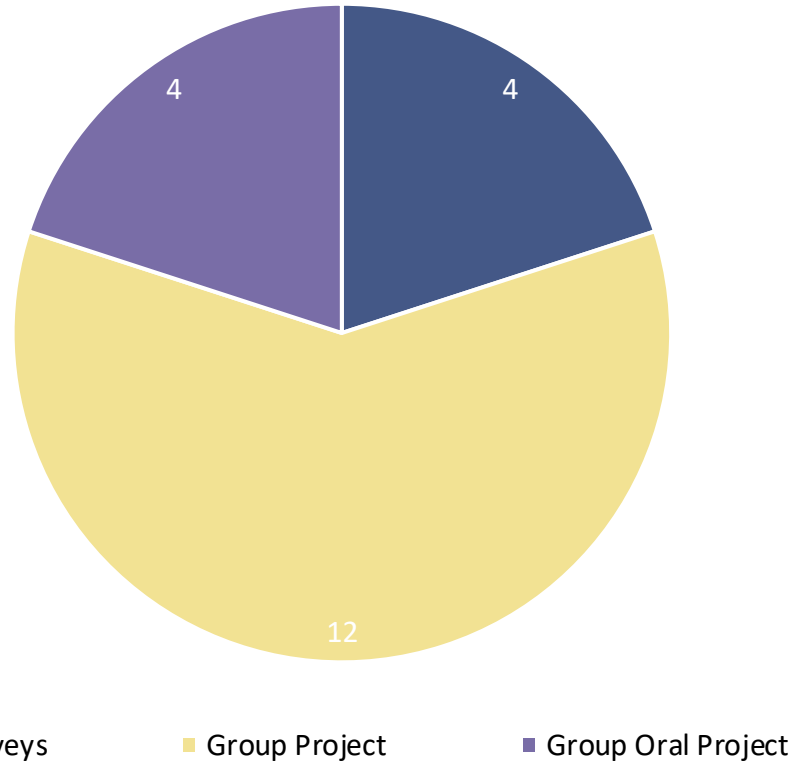
Chapter	Sections
01: What's Big Data	○ Introduction to Big Data ○ Introduction to No SQL ○ Understand Hadoop framework ○ Understand The Data Engineering Lifecycle
02: Get Started with Databricks Data Science & Engineering Workspace	○ Understand Spark ○ Understand the core components of the Databricks Lakehouse platform ○ Use PySpark in Databricks
03: Transform Data with Spark	○ Extract data from a variety of file formats and data sources using Spark ○ Apply a number of common transformations to clean data using Spark ○ Reshape and manipulate complex data using advanced built-in functions in Spark ○ Leverage UDFs for reusable code and apply best practices for performance in Spark
04: Manage Data with Delta Lake	○ What is Delta Lake ○ Manipulate Delta table through PySpark
05: Work with data warehouses	○ Analyze data in a relational data warehouse ○ Load data into a relational data warehouse

Summary

Chapter	Sections
Annexe: Cloud Computing Introduction	○ Cloud Computing: Basics concepts ○ Scalable & Available ○ Cloud-native toolkit
Annexe: AWS & Azure Introduction	○ AWS : Basics concepts ○ Interact with AWS ○ Use Amazon S3 ○ Azure : Basics concepts

Mark


IT'S YOUR TURN TO SHINE!!


4
4
12
Surveys
Group Project
Group Oral Project